Karan Gupta

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.48
Capt. Ram Singh Degree College	Class XII (CBSE)	2023	94.20%
S M P K D Shikshan Sansthan	Class X (CBSE)	2021	92.70%

SCHOLASTIC ACHIEVEMENTS

JEE Advanced 2023: Secured **AIR 1026**, placing among the **top 0.4% nationwide**.

JEE Main 2023: Secured **AIR 1230**, placing among the **top 0.1% nationwide**.

IICPC QuantFest 2025: Qualified for the **Regional Round**, securing **AIR 327 among 10,000+ participants**.

Codeforces: Achieved **Specialist** rating with a **peak rating of 1483**.

IIT Pokerbots: Secured **AIR 99 among 5,000+ participants**.

Optiver Trade-A-Thon: Secured **AIR 15 among 1,800+ participants** across different IITs.

WORK EXPERIENCE

Software Engineer Intern, Atlassian

- **Architected and delivered Site Optimizer**, providing actionable Jira configuration insights that reduce **200+ hours of manual analysis annually per administrator** across large-scale enterprise instances.
- **Developed Auto Conflict Resolver (ACR)**, an AI-assisted merge conflict automation framework projected to save **~$910K annually** by automatically resolving safe conflicts while retaining human review for high-risk changes.
- **Extended Site Optimizer** with support for **Custom Field Options and Issue Security Level Schemes**, using a lightweight **guardrails table refreshed every 24 hours** to improve query performance while maintaining decision accuracy.

Founding Software Engineer, Solve Arena

- Architected and scaled an AI-powered competitive learning platform serving **12,000+ active users**, owning the full-stack architecture from backend infrastructure to deployment.
- Led end-to-end product development, delivering real-time contests, AI-powered learning features, and scalable cloud infrastructure that drove platform growth.

PROJECTS

Reputation-Weighted BFT Leader Endorser Selection (MPI, Hyperledger Fabric, BFT)

- Developed a **reputation-based leader selection framework** for Hyperledger Fabric using TP, TN, FP, FN metrics with FN-aware penalties and Laplace smoothing.
- Implemented **epoch-based BFT leader election and an MPI-based distributed simulator**, achieving **99.8% successful commits** with the Byzantine peer elected 0 times in a 500-transaction evaluation.

Bayesian Estimation & MCMC Methods for Financial Time Series (Bayesian Statistics, MCMC, Python)

- Modeled **XRP/USDT log returns** using Bayesian Normal-Inverse-Gamma inference and **rejection sampling**, generating posterior estimates and predictive intervals.
- Implemented **Metropolis-Hastings and Gibbs sampling** with Bayes factor analysis, with Gibbs producing approximately **50× more effective samples**

JADE: A Domain Specific Language for Distributed Computing (C++, Lex, Yacc)

- Developed **JADE**, a declarative DSL and compiler for parallel/distributed computing, implementing **lexical analysis, parsing, semantic analysis, and C++/MPI code generation** with automatic role-to-rank mapping.
- Implemented **compile-time communication validation and deadlock detection**, automatically translating high-level primitives such as broadcast, send, recv, and gather into optimized MPI operations while eliminating manual rank and communicator management.

gemOS Virtual Memory Management (C, Operating Systems)

- Designed and implemented **mmap, munmap, and mprotect** by managing per-process VM-area linked lists, with **lazy allocation and copy-on-write** support through a custom cfork() syscall and page fault handler.
- Improved **memory efficiency and process forking** with minimal overhead, mirroring modern virtual memory management techniques. Implemented as part of the **Operating Systems II** course.

CCIRP: Central Communication and Intelligent Reminder Platform (Next.js, MongoDB, FastAPI, Redis, Kafka)

- Built a **multi-channel communication platform** for email, SMS, and WhatsApp campaigns with scheduling, audience management, and analytics.
- Implemented **asynchronous dispatch, engagement tracking, recipient segmentation, and a Gemini-powered assistant** for centralized communication.

RELEVANT COURSES

Core CS: DSA, Operating Systems, DBMS, Computer Architecture, Software Engineering, Theory of Computation, Compiler, Computer Networks, Distributed Computing, Parallel and Concurrent Programming, Approximation Algorithms

AI/ML: Intro to AI, Intro to Modern AI, Foundation of ML, Deep Learning, Fraud Analytics, Practical Malware Awareness

Mathematics: Discrete Mathematics, Linear Algebra, Probability, Statistics, Calculus I/II, Transform Techniques, Bayesian Statistics, Linear Optimization, Convex Optimization

POSITIONS OF RESPONSIBILITY

Technical Manager, Office of Career Services (OCS), IIT Hyderabad

- Own the development and maintenance of the institute's placement & internship portal, serving **1,500+ students**.
- Build features that streamline recruiter onboarding, student applications, and hiring workflows.

Head of Software Domain, *Tinkerers' Lab*

- Led full-stack development across 5+ internal projects and Summer School initiatives, impacting 50+ participants.
- Mentored 10+ junior members and delivered project pipelines under strict academic timelines

SKILLS

Core CS Skills

- Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Database Management, Computer Networks, Distributed Computing

Technical Skills

- *Programming Languages***:** C/C++, Go, Python, Java, JavaScript, Bash
- *Software Development***:** Full-Stack Development, Backend Development, Frontend Development, REST APIs, Multi-threading
- *AI/ML***:** Machine Learning, Deep Learning, Transformers, Large Language Models (LLMs), Generative AI, **RAG, GANs, NLP**
- *Frameworks & Technologies***:** React, Next.js, Node.js, Express.js, FastAPI, Flutter, Kafka, Redis, Celery
- *Databases & Tools***:** PostgreSQL, MySQL, MongoDB, Firebase, Git, GitHub, Linux